

Thorstein Veblen and post-Darwinian economics

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1. Introduction

Before the end of the 19th century, Thorstein Veblen ([1898A], 1919, p. 56) asked in a famous article: 'Why is economics not an evolutionary science?' The term 'evolutionary' was subsequently adopted by institutional economists, but often in broad or developmental terms, and with only slight attention to the more precise mechanisms of natural selection as developed in biology. Veblen's knowledge of biological science was remarkably up-to-date, yet evolutionary theory has developed enormously since his death in 1929, leaving many of his institutionalist followers well behind. Veblen made a direct appeal to biological science for inspiration; but subsequently, and until very recently, this example has rarely been replicated. Accordingly, there has been remarkably little detailed exploration, informed by biology, of what Veblen precisely meant by an 'evolutionary' science, and of the character of the 'post-Darwinian' economics that he attempted to build.¹

The idea that economic development is in some sense evolutionary has a long pedigree. For instance, Karl Marx (1977, p. 525) saw Charles Darwin's theory as 'a natural-scientific basis for the class struggle in history'. Alfred Marshall (1949, p. xii) stated that 'the Mecca of the economist lies in economic biology rather than in economic dynamics'. And Joseph Schumpeter (1976, p. 82) insisted that the 'essential point to grasp is that in dealing with capitalism we are dealing with an evolutionary process . . . a fact . . . long ago emphasized by Karl Marx'.²

Despite these statements, the application of biological ideas to economics remains limited, and the principal influence on economics from the natural sciences is still that of 19th-century physics (Hamilton, 1991; Ingrao and Israel, 1990; Mirowski, 1989). In the hands of a minority of researchers, however, evolutionary ideas in economics have

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¹ A sympathetic and knowledgeable adversary such as Frank Knight (1935, p. 209) responded to Veblen's assertion that he had applied Darwinian principles to economics with the riposte: 'I cannot find anything in particular that he could reasonably have meant by this claim'. More recently, Lamar Jones (1986) argued that Veblen's attachment to Darwinism involved 'a case of mistaken identity'. Both standpoints are rebutted by the argument in this essay.

² For a discussion of the conception of evolution in Marx's writings see Hodgson (1992, 1993). In Hodgson (1993) I argue that Veblen's use of evolutionary thinking from biology was much more extensive than that of Schumpeter. For Marshall's attitude to evolutionary biology see (Hodgson, 1993; Thomas, 1991).

enjoyed a remarkable revival in the 1980s. In this context, it is all the more appropriate that Veblen's works should be re-examined.

Like Marshall, Veblen saw that the appropriate metaphor for economics was to be found in biology. In particular, Veblen saw the evolutionary metaphor as crucial to the understanding of the processes of technological development in a capitalist economy. But unlike his English colleague, he did not care to develop a static, equilibrium analysis as a prelude to the dynamic. He characterised his own economics as post-Darwinian, and argued that economics should embrace the metaphor of evolution and change, rather than the static ideas of equilibrium that had been borrowed by the neoclassical economists from physics.

However, after rebutting a mechanical prelude to economic dynamics, Veblen was faced with a biology at a state of development at which the mechanisms of evolution were only partly understood. Consequently, and given his own personal aversion to intellectual 'symmetry and system-making' (Veblen, [1899], 1919, p. 68), there was little chance that Veblen would be able to build an economic theory on the Marshallian scale.

Instead, he leaves us with plentiful hints and insights, many of which are brilliant, several contradictory. He writes in a style which is often dazzling and illuminating, but also sometimes evasive or unclear. Partly for the latter reason, and partly because he did not provide us with a systematic theoretical legacy, his significance for evolutionary economics in particular, and economics in general, remains underestimated to this day.

Importantly, Veblen had a keen and perceptive understanding of the relationships and connections between the social and the physical sciences. In this essay it is shown that Veblen had two primary reasons for the adoption of a Darwinian and evolutionary metaphor. One relates to the idea of cumulative causation and an opposition to depictions of the economic process that are consummated in equilibrium. The other is based on the formation of analogies to both the gene and the processes of natural selection in the social world.

Darwinian natural selection involves several component principles. First, there must be sustained variation among the members of a species or population. Variations may be blind, random, or purposive in character, but without them, as Darwin insisted, natural selection cannot operate. Second, there must be some principle of heredity or continuity, through which offspring have to resemble their parents more than they resemble other members of their species. In other words, there has to be some mechanism through which individual characteristics are 'passed on' through the generations. Third, natural selection itself operates either because better-adapted organisms leave increased numbers of offspring, or because the variations or gene combinations that are preserved are those bestowing advantage in struggling to survive. The latter is the principle of the struggle for existence. It is important to note that evolutionary selection in biology occurs both by differential rates of death and differential rates of birth; it is a matter of procreation as well as destruction. The application of the metaphor of natural selection to economics must be on the basis of analogous principles. It is argued here that Veblen was relatively successful in this regard.

Section 2 of this essay discusses the origins and development of Veblen's evolutionary thinking, with comparisons with the work of Herbert Spencer and others. In Section 3 Veblen's evolutionary theory is probed more deeply, relating it to his idea of cumulative causation and the differing conceptions of Darwin and Lamarck. Section 4 concludes the essay.

2. The genesis of Veblen's evolutionary thought

As a student at Yale University, Veblen came under the influence of William Graham Sumner, the Social Darwinist (Dorfman, 1934, pp. 43–46; Riesman, 1963, p. 19). In addition, Veblen read widely, including biology, psychology and philosophy, as well as the social sciences. Significantly, the second article that Veblen published, in 1892, was on Herbert Spencer, and was 'offered in the spirit of the disciple' (Veblen, [1892], 1919, p. 387). This was at a time when Spencer's writings were as popular as those of Darwin, and to some extent, as Peter Bowler (1983) explains, Darwinism was even in eclipse. As a result of these contemporary influences, Veblen became immersed in evolutionary theory, and clear, enduring traces of both Darwin and Spencer can be found in his thought.¹

Veblen's awareness of developments in evolutionary biology was sufficiently up-to-date for him to refer to principles of Mendelian genetics in his work (Veblen, 1914, pp. 21–25; 1915, pp. 277–278; 1919, pp. 457–473). This was only shortly after they were beginning to be accepted by some biologists as a possible explanation of variation and heredity in evolutionary theory, and decades before the flowering of the neo-Darwinian hybrid formed by growing Mendelian genetics on a Darwinian stem. He was also acquainted with the mutation theory of Hugo de Vries (Veblen, 1919, pp. 462–467).

It appears that under the sway of these thinkers, as well as leading pragmatist philosophers such as Charles Sanders Peirce and William James, Veblen attempted to develop a theory of socioeconomic evolution. Presuming that human behaviour is dominated by habits of thought, Veblen enquired as to the causes of these habits. Being influenced by the instinct theory in Spencer's *Principles of Psychology*, as well as William McDougall's *Introduction to Social Psychology*, Veblen took the view that habits were partly rooted in such instincts. As in Spencer's work, they were seen to represent evolutionary adaptations to changing environmental conditions. However, Veblen identified instincts of both a progressive and regressive kind, and stressed the conflict between them in the modern world.

Instincts and institutions

From the concept of habits of thought, it was a short step for Veblen to a definition of institutions. According to him, institutions are 'settled habits of thought common to the generality of men' (Veblen, [1909], 1919, p. 239). In other words, they are seen as an outgrowth of the routinised thought processes that are shared by a number of persons in a given society.²

Crucially, the presumed nature and origin of the different features of instincts and habits in Veblen's work is unclear. In many places he seems to regard their diversity as having racial origins. In others he hints at a sophisticated idea of a conflict between

¹ See Dorfman (1934, p. 46), Edgell (1975), Edgell and Tilman (1989), Eff (1989), Murphree (1959), Peel (1971, p. 198). For a good general discussion of the origins of American institutionalism see Mayhew (1987).

² Note the famous definition of the institution by the later institutionalist writer, Walton Hamilton (1963, p. 84): 'It connotes a way of thought or action of some prevalence and permanence, which is embedded in the habits of a group or the customs of a people . . . Institutions fix the confines of and impose form upon the activities of human beings.' This may be preferred to Veblen's rather mentalistic definition because of its inclusion of human action as well as thought.

Nevertheless, in the writings of the American pragmatists who influenced Veblen, there is a close connection between habits of thought and action. For instance, Charles Sanders Peirce (1878, p. 294) wrote: 'The essence of belief is the establishment of habit; and different beliefs are distinguished by the different modes of action to which they give rise . . . the whole function of thought is to produce habits of action'. This identification of belief with habit, which is manifest in action, is a cornerstone of Peirce's pragmatism, and to some extent explains Veblen's terminology.

dissimilar institutional developments, each one locked-in to separate tracks. But, as we have noted, this lack of sufficient explanation, alongside brilliant insight, is one of the frustrating and characteristic features of Veblen's writing.

As Donald Walker (1977) contends, Veblen's account of the relationship between instincts, habits of thought and social institutions is not entirely coherent. In response, Malcolm Rutherford argues that Veblen started from a genetic research programme with reductive aspects:

He attempted to follow a causal chain from a pre-institutional state of affairs to the America of the twentieth century, and to do so he required some non-institutional, exogenously determined human drives with which to start the analysis. However, given the rejection of orthodox psychologism, he also had to argue that these instinctive drives could easily be overturned or contaminated in the course of institutional development (Rutherford, 1984, p. 333).

Veblen may have originally entertained a reductionist position in which explanations of human behaviour can be reduced to instinctive drives. However, he quickly moved away from it when he realised that institutions could be seen as not only being formed by, but formative of, such elements.

This development and change in his thought accounts for some of its contradictory aspects. Veblen often repeats that habits of thought are moulded or even inculcated by culture, practice or technology. For example: 'What the discipline of the machine industry inculcates . . . in the habits of life and thought of the workman, is regularity of sequence and mechanical precision' (Veblen, 1904, p. 309). More generally: 'A habitual line of action constitutes a habitual line of thought, and gives the point of view from which facts and events are apprehended and reduced to a body of knowledge' (Veblen, [1898B], 1934, p. 88). Thus, for Veblen, habits of thought are founded not simply on the instincts but on human culture and habitual action.

Furthermore, in at least one place, and especially in his later works, Veblen (1914, pp. 2, 7) made it clear that instincts are not purely biological, nor genetically transmitted. Indeed, not only were instincts seen to give rise to habits of thought but they were sometimes described as being formed by them. Instinct is thereby 'a matter of tradition out of the past, a legacy of habits of thought accumulated through the experience of past generations.' It 'falls into conventional lines, acquires the consistency of custom and prescription, and so takes on an institutional character and force' (Veblen, 1914, p. 7).

Notwithstanding this, and as Charles Leathers (1990, p. 166) notes, Veblen still failed to explain how instincts 'became and remain heredity elements of human nature.' Although Darwin, Spencer and Sumner put Veblen on this path, his 'use of instinct theory declined markedly in his later work, and for most of his positive theory of institutions and institutional change his instinct theory has only minor significance' (Rutherford, 1984, p. 313).

Nevertheless, the incomplete move against biological or genetic reductionism in Veblen's work is of enormous significance. It distinguishes him from the prevailing biologism of the late 19th century as well as from the ubiquitous evolutionary theory of Herbert Spencer. Veblen (1899, p. 188) clearly saw institutions as well as individuals as units of evolutionary selection. With modern hindsight, this suggests the notion that the information transmitted through learning or imitation to institutions or individuals was analogous, but different from, the transmission of genetic information in the process of biological evolution. Consequently, institutions are both replicators and the units of selection in socioeconomic evolution. In any case, Veblen's adoption of institutions as units of selection clearly demarcates his theory from that of Spencer.

Veblen and Spencer

Veblen noted several criticisms of Spencer's theory in his works.¹ Other objections, like the final point in the last paragraph, were implicit. Whilst Spencer anticipated a capitalist system in which peaceful industrial activity excluded militarism and war, Veblen, in contrast, regarded the pecuniary activities of the 'absentee owners', or shareholders, as essentially dysfunctional and predatory in character. For Veblen, as for Marx, capitalism would always have inner conflicts and imperfections. Accordingly, Veblen argued that Spencer's typology of social systems was not exhaustive and that the option of socialism provided an escape from its false alternatives (Veblen, [1892], 1919, pp. 402–405).

Second, Veblen had a much more adequate explanation of the sources of creativity and variety in socioeconomic evolution. In some passages Veblen relied on biological mutation. In these instances he shared Spencer's biological reductionism but his theory contrasted with that of Spencer by incorporating the ideas of Johann Gregor Mendel. Elsewhere, Veblen (1914) devised the concept of 'idle curiosity' and this can serve as a genesis for diversity and variation. He suggested that the human tendency towards experimentation and creative innovation could generate novelty in an ongoing manner.² This could lead to new and improved ways of thinking and doing, and consequently the generation of the greater variety upon which evolutionary selection would operate. For Veblen, in particular, 'idle curiosity' is a major source of technological change. In other passages the principle of cumulative causation was itself relied upon to explain cumulative divergence, potential instability and novelty.

Third, as in his criticisms of Marx, Veblen rejected the notion of the perfectibility of society. The same critique applied to Spencer's ideal state of *laissez-faire*. Being more a Darwinian than Spencer, Veblen inclined to the Malthusian critique of the notion that society could be perfected or human happiness optimised. As Robert Young argues, Spencer's belief in progress and perfectibility

was more reminiscent of the doctrines that prompted Malthus to write the polemical first edition of his *Essay on the Principle of Population*. Spencer's *laissez faire* optimism is far closer to Rousseau, Condorcet and Godwin than to Malthus (Young, 1969, p. 137).

Thus Spencer's evolutionary theory was fundamentally anti-Malthusian, at least in this respect.³ Although Veblen had socialist learnings, he argued, in contrast, against the idea of finality or consummation in economic development. Variety and cumulative causation mean that history has 'no final term' (Veblen, [1908B], 1919, p. 37). In Marxism 'the final term was socialism or the classless society. Veblen rejected the teleological concept of a final goal as pre-Darwinian' (Hill, 1958, p. 138).

Notably, Malthus saw diversity as both initial to change and renewed through time, and hinted at a process of selection that was to inspire Darwin. Like Lamarck, Spencer saw the source of variety in the adaptation of individuals to their environment (Burkhardt, 1977).

¹ See, for example, Veblen ([1892], 1919, pp. 402–405; [1908A], 1919, p. 192n.).

² Note that a similar idea was put forward by John A. Hobson—the closest contemporary English thinker to the American institutionalists—in a work published in the same year as Veblen's *Instinct of Workmanship*. Hobson (1914, pp. 240–241, 336) saw the role of human error and playful inventiveness as decisive in creating mutations in behavioural patterns, and thereby a source of continuous evolutionary innovation. Interestingly, he gave more stress to the function of the 'freedom of the human will' than Veblen in this context. However, Hobson did not go so far as Veblen to incorporating this idea into an evolutionary theory of the phylogenetic type.

³ In addition, and in contrast to Veblen, Spencer's writings had a strong Panglossian tone. For a critique of Panglossian applications of evolutionary theory see Hodgson (1991B).

Organisms adjust continuously as if attempting to reach harmony with their surroundings. For Lamarck and Spencer, the environment was the key agent of change. It was for Darwin too, but for him 'variation was present first, and the ordering activity of the environment ('natural selection') followed afterwards' (Mayr, 1982, p. 354). For Darwin, change resulted from a combination of variation and environmental selection. It was this insight—which Darwin had taken and modified from Malthus—that Veblen followed. Hence the fourth major difference is that Veblen and Spencer have different conceptions of the place and function of variety in the evolutionary process.

Fifth, Veblen rejected Spencer's rationalistic and hedonistic psychology:

Spencer is both evolutionist and hedonist, but it is only by recourse to other factors, alien to the rational hedonistic scheme, such as habit, delusions, use and disuse, sporadic variation, environmental forces, that he is able to achieve anything in the way of genetic science, since it is only by this recourse that he is enabled to enter the field of cumulative change within which the modern post-Darwinian sciences live and move and have their being (Veblen, [1908A], 1919, p. 192n.)

Accordingly, there is a considerable divergence between the ideas of Veblen and Spencer.

3. Veblen, Darwinism and cumulative change

The significance of the last quotation given above needs to be elaborated, for it contains a key to Veblen's adoption of a 'post-Darwinian' approach.¹ Veblen ([1908B], 1919, p. 37) argues that pre-Darwinian science was 'taxonomic':

To the pre-Darwinian taxonomists the center of interest and attention, to which all scientific enquiry must legitimately converge, was the body of natural laws governing phenomena under the rule of causation. These natural laws were of the nature of rules of the game of causation. They formulated the immutable relations in which things 'naturally' stood to one another before causal disturbance took place between them, the orderly unfolding of the complement of causes involved in the transition over this interval of transient activity, and the settled relations that would supervene when the disturbance had passed and the transition from cause to effect had been consummated—the emphasis falling on the consummation.

In opposition to this 'taxonomic' view, Veblen ([1908B], 1919, p. 37) writes:

The characteristic feature by which post-Darwinian science is contrasted with what went before is a new distinction of emphasis, whereby the process of causation, the interval of instability and the transition between initial cause and definitive effect, has come to take the first place in the enquiry; instead of that consummation in which causal effect was once presumed to come to rest. This change in point of view was, of course, not abrupt or catastrophic. But it has latterly gone so far that modern science is becoming substantially a theory of the process of consecutive change, realized to be self-continuing or self-propagating and to have no final term (Veblen, [1908B], 1919, p. 37).

It should be noted, however, that for Veblen post-Darwinian biology does not transcend mechanical analogies:

in the Darwinian scheme of thought, the continuity sought in and imputed to the facts is a continuity of cause and effect. It is a scheme of blindly cumulative causation, in which there is no trend, no final term, no consummation. The sequence is controlled by nothing but the *vis a tergo* of brute causation, and is essentially mechanical (Veblen, [1907], 1919, pp. 436).

Thus for Veblen a key attraction of Darwinian ideas is more to do with the examination of causal processes, of a never-ending and cumulative nature, never reaching a state of

¹ It has been suggested by Lamar Jones (1986, p. 1053) that for Veblen part of the appeal of Darwin's ideas is due to the fact that they were perceived to undermine the oppressive intolerance of contemporary religion by promoting a naturalistic or materialistic world-view. The truth or untruth of this is not easily established, but Jones fails to note the additional and manifest significance of Veblen's interpretation of Darwinism based on 'cumulative causation', as elaborated here.

equilibrium or rest. Their lure is not to do with any supposed alternative to mechanistic theory.

Limitations of orthodox economic theory

The emphasis on cumulative change is a persistent theme in Veblen's work, signalling a source of fundamental divergence between his type of thinking and the equilibria-orientated conceptions of both neoclassical economics and Spencerian theory. Criticisms in this vein are raised, for example, against the work of Marshall, whose

theorems apply to the various phases of the development not as accounting for the developmental sequence, but as limiting the range of variation. They say little, if anything, as to the order of succession, as to the derivation and the outcome of any given phase, or as to the causal relation of one phase of any given economic convention or scheme of relations to any other. They indicate the conditions of survival to which any innovation is subject, supposing the innovation to have taken place, not the conditions of variational growth (Veblen, [1900], 1919, pp. 176–177).

Similarly, Veblen ([1898A], 1919, p. 70) criticises the economics of Carl Menger and the Austrian School: The 'Austrian group struck out on a theory of process, but presently came to a full stop because the process about which they busied themselves was not, in their apprehension of it, a cumulative or unfolding sequence.'

In the work of such economists, Veblen argues, the process stops for a simple reason: a 'faulty conception of human nature.' He elaborates:

In all received formulations of economic theory, whether at the hands of English economists or those of the Continent, the human material with which the inquiry is concerned is conceived in hedonistic terms; that is to say, in terms of a passive and substantially inert and immutably given human nature (Veblen, [1898A], 1919, p. 73).

Hence the cumulative quality of evolutionary thinking rests precisely on the fact that it does not take human nature or preference functions as given or for granted.¹ Both the circumstances and temperament of an individual are part of the cumulative processes of change:

They are the products of his hereditary traits and his past experience, cumulatively wrought out under a given body of traditions, conventionalities, and material circumstances; and they afford the point of departure for the next step in the process (Veblen, [1898A], 1919, pp. 74).

Veblen goes on to recognise, most perceptively, that both the individual and the environment, in which selection takes place, are affected by these unceasing developments:

The economic life history of the individual is a cumulative process of adaptation of means to ends that cumulatively change as the process goes on, both the agent and his environment being at any point the outcome of the last process (Veblen, [1898A], 1919, pp. 74–75).

Clearly, marginalist theory is limited in this respect:

To any modern scientist interested in economic phenomena, the chain of cause and effect in which any given phase of human culture is involved, as well as the cumulative changes wrought in the fabric of human conduct itself by the habitual activity of mankind, are matters of more engrossing and more abiding interest than the method of inference by which an individual is presumed invariably to balance pleasure and pain under given conditions that are presumed to be normal and invariable (Veblen, [1909], 1919, p. 240).

¹ It should be noted that in some passages Marshall (1949, pp. 1, 76) also accepted that individual wants are moulded by circumstances. For discussions of this see Parsons (1937, ch. 4), Chasse (1984). But this is not the Marshall that is formalised into theorems, or represented in the textbooks.

What Veblen was seeking was precisely a theory as to why change and transformation takes place, not a theory which muses over equilibrium conditions after individual preferences and technological possibilities are established. 'The question', he wrote, 'is not how things stabilise themselves in a "static state", but how they endlessly grow and change' (Veblen, [1925], 1934, p. 8). In sum, as Veblen ([1898A], 1919, p. 77) asserts:

an evolutionary economics must be a theory of a process of cultural growth as determined by the economic interest, a theory of a cumulative sequence of economic institutions stated in terms of the process itself.

This passage is one of many where Veblen presents the idea of cumulative causation. Veblen's work in this area is thus an important precursor to that of Allyn Young (1928), Gunnar Myrdal (1939, 1944, 1957), Nicholas Kaldor (1972) and K. William Kapp (1976). Despite Veblen's earlier discussion, the idea of cumulative causation is more frequently associated with Myrdal and Kaldor.

A brief discussion of these Veblen-Myrdal and Veblen-Kaldor connections is appropriate. Myrdal originally took the idea of cumulative causation from the monetary economics of Knut Wicksell, coming to regard himself as an institutionalist later, in the 1940s.¹ However, Kaldor was taught and inspired by Young, and it is almost certain that Young, in turn, was prompted by Veblen's repeated use of the 'cumulative causation' idea when he associated it with the particular phenomenon of increasing returns (Young, 1928). Young, 'a friend of Mitchell and admirer of Veblen' (Dorfman, 1964, p. xlviii) taught Kaldor, and hence there is a clear—but hitherto unexplored—link between the Veblenian and Kaldorian schools of thought.

Institutional inertia

It is suggested that cumulative causation can occur at all levels of social evolution, from the formation of institutions to macroeconomic development. Thus Veblen ([1909], 1919, p. 241) argues that:

institutions are an outgrowth of habit. The growth of culture is a cumulative sequence of habituation, and the ways and means of it are the habitual response of human nature to exigencies that vary incontinently, cumulatively, but with something of a consistent sequence in the cumulative variations that so go forward.

The cumulative, self-reinforcing aspect of routinised behaviour and social institutions is highlighted in a work as early as *Theory of the Leisure Class*:

The situation of today shapes the institutions of tomorrow through a selective, coercive process, by acting upon men's habitual view of things, and so altering or fortifying a point of view or a mental attitude handed down from the past At the same time, men's present habits of thought tend to persist indefinitely, except as circumstances enforce a change. These institutions which have so been handed down, these habits of thought, points of view, mental attitudes and aptitudes, or what not, are therefore themselves a conservative factor. This is the factor of social inertia, psychological inertia, conservatism (Veblen, 1899, pp. 190–191).

It would seem that the cumulative and self-reinforcing aspect of institutions and routines relates to some kind of process of positive feedback. In this respect there is another contrast with orthodox economics, in which the formation of equilibrium relies upon negative feedback processes, such as diminishing returns to scale. Rather than equilibrium, positive feedback can engender such phenomena as lock-in (to use the

¹ For a discussion of cumulative process models from Thornton to Wicksell see Humphrey (1990). See Myrdal (1978) for an account of his conversion to institutionalism.

modern parlance), where outcomes become frozen because of their self-reinforcing attributes (Arthur, 1989, 1990). Such locked-in phenomena can thus be regarded as sufficiently stable units of selection in an evolutionary process.

However, cumulative reinforcement of a number of parallel institutions can eventually lead to conflict and disruption. Stephen Edgell (1975, pp. 272–273) summarises Veblen's view in these terms: 'institutions that emerge during one era may persist into another and the resulting cultural lag is likely to give rise to "friction" between the habits of thought generated by the new material conditions and the habits and institutions more appropriate to an earlier period of cultural development'. The processes underlying institutional change can be likened to strata shifting slowly at different rates, but occasionally causing seismic disturbance and discontinuities.

Again to put this Veblenian notion in a modern context, W. Brian Arthur (1990, p. 85) notes that the outcomes of conflicts between several simultaneously locked-in processes can give rise to an idea of discontinuous development akin to the idea from modern biology of 'punctuated equilibria' (Eldredge and Gould, 1977; Gould, 1982).¹

Institutions as units of selection

Furthermore, by recognising the durable character of institutions, Veblen thus discovered an equivalent to the gene in the socioeconomic world. This is the second main reason for his adoption of the metaphor of Darwinian evolution. Socioeconomic evolution is regarded as a selection process, working on institutions as replicators and units of selection, combined with the simultaneous processes of adaptation of both individuals and institutions to their mutual environment.²

The life of man in society, just as the life of other species, is a struggle for existence, and therefore it is a process of selective adaptation. The evolution of social structure has been a process of natural selection of institutions. The progress which has been and is being made in human institutions and in human character may be set down, broadly, to a natural selection of the fittest habits of thought and to a process of enforced adaptation of individuals to an environment which has progressively changed with the growth of community and with the changing institutions under which men have lived. Institutions are not only themselves the result of a selective and adaptive process which shapes the prevailing or dominant types of spiritual attitude and aptitudes; they are at the same time special methods of life and human relations, and are therefore in their turn efficient factors of selection. So that the changing institutions in their turn make for a further selection of individuals endowed with the fittest temperament, and a further adaptation of individual temperament and habits to the changing environment through the formation of new institutions (Veblen, 1899, p. 188).

There are many interesting facets of this passage, which space does not permit us to dwell upon here. For instance, Veblen does sometimes seem to equate evolution with progress, as Lamar Jones (1986, p. 1053) has noted. However, this was the almost universal disposition of both biological and social evolutionary thinkers until quite recently (Nitecki, 1988). And further, Veblen ([1908B], 1919, p. 55) seems to take a different view when he later criticises adepts of evolutionary thought for infusing their science with a 'beneficent trend; so that "evolution" is conceived to mean amelioration or "improvement"'. He also considers 'spectacular instances of the triumph of imbecile institutions over life and culture' (Veblen, 1914, p. 25).

¹ The use of the word 'equilibria' by these biologists should not mislead. In this context the term is different from that found in mechanics and orthodox economics, in that to some degree it encompasses mutation and change within the same evolutionary track. Furthermore, the punctuations refer to abrupt speciation and are just as important as the equilibria. For further discussions of punctuated equilibria in a social context see Mokyr (1990, forthcoming) and Somit and Peterson (1989).

² For a discussion of different interpretations of Veblen's theory of institutional change see Tilman (1987).

What is most interesting is Veblen's suggestion that institutions and habits of thought are units of selection in an evolutionary process. Whilst these are more malleable and do not mutate in the same way as their analogue in biology, institutions and settled habits do have a sufficient degree of durability to regard them as having quasi-genetic qualities. Veblen thus implied that biology and economics were on different ontological levels, but united by some broad and common evolutionary themes.¹

Not only does this idea provide Veblen with a biological metaphor, it also becomes a basis for analysis of fundamental economic activities. Work, for instance, is seen as involving a degree of practical knowledge or know-how which is both acquired and routinised over time. Indeed, the industrial skill of a nation consists of a set of relevant habits, acquired over a long time, widely dispersed through the employable work force, reflective of its culture and deeply embedded in its practices (Veblen, 1914; Dyer, 1984). In the last quarter of the 20th century this idea has been rediscovered by Richard Nelson and Sidney Winter (1982) and applied to their evolutionary theory of the firm.

Veblen turns his conception of the quasi-genetic quality of habits of thought into a 'Darwinian' theory of the relationship between ideas and vested interests. This clearly demarcates him from all major alternative streams of economic thought. As the following quotation shows, he breaks explicitly from Marx on this point:

Under the Darwinian norm it must be held that men's reasoning is largely controlled by other than logical, intellectual forces; that the conclusion reached by public or class opinion is as much, or more, a matter of sentiment than of logical inference; and that the sentiment which animates men, singly or collectively, is as much, or more, an outcome of habit and native propensity as of calculated material interest. There is, for instance, no warrant in the Darwinian scheme of things for asserting *a priori* that the class interest of the working class will bring them to take a stand against the propertied class (Veblen, [1907], 1919, p. 441).

Note that there is also here a clear implicit contrast with optimising rationalism of neo-classical economics, and a rebuttal of the kind of faith in persuasion and reason to be found later in the writings of John Maynard Keynes.²

Veblen suggests that whilst Marx saw the importance of human action, the assumed basis of its motivation reduced to the problematic notion of 'class interest'. This ignores the processes through which perceptions of circumstances are *interpreted* and thereby transformed into human drives and purposes. Veblen saw institutions and 'habits of thought' as providing the framework for such a cognitive interpretation. In general, Veblen's adherence to a 'Darwinian' theory involved a meticulous examination of the causal mechanisms involved in human behaviour, rather than assumptions of either ad hoc maximising propensities or vague 'social forces'.

Purposeful behaviour and Lamarckism

It is sometimes alleged that Veblen 'teeters between free will and determinism' (Seckler, 1975, p. 56), and entertained a conception of science from which purpose and intentionality were excluded (Commons, 1934, p. 654). Nevertheless, Veblen's emphasis on human

¹ The question is raised here of the appropriate unit(s) of selection in socioeconomic evolution. On this see Hodgson (1991A) and the references therein.

² Forest Hill (1958, p. 139) elaborates Veblen's critique of Marx with the following words: 'In Veblen's opinion, Marx uncritically adopted natural rights and natural law preconceptions and a hedonistic psychology of rational self-interest. On these bases Marx elaborated his labor theory of value, with labor as the source and measure of value, and the corollary doctrines of labor's right to its full product, of surplus value, and exploitation of labor. He attributed rational self-interest not only to individuals but to entire classes, thereby explaining their asserted solidarity and motivation in class struggle. Veblen rejected the concept of rational class interest and the labor theory of value, along with its corollaries and natural rights basis.'

purposefulness is repeated and significant: 'Economic action is teleological, in the sense that men always and everywhere seek to do something' (Veblen, [1898A], 1919, p. 75).

Veblen saw human purposeful behaviour itself as the result of natural selection. Regarding 'man':

Like other species, he is a creature of habit and propensity. But in a higher degree than other species, man mentally digests the content of habits under whose guidance he acts, and appreciates the trend of these habits and propensities . . . By selective necessity he is endowed with a proclivity for purposeful action (Veblen, [1898B], 1934, p. 80).

Nevertheless, the nature and definition of what is meant by 'purposeful action' is not always clear, and this is perhaps one source of controversy over Veblen's line of thought (Seckler, 1975; Langlois, 1989). However, Veblen's emphasis on 'idle curiosity', for instance, as well as many of his own explicit statements, would seem to retain a crucial role for human agency in his theory. On the other hand, Veblen held onto a traditional, 19th century view of science which rejected all purpose in the investigation of the facts.

Whatever the appropriate interpretation, any explicit emphasis on purposeful behaviour creates a problem for a strictly 'Darwinian' account of socioeconomic evolution. Purposeful behaviour is more often associated with evolution of a Lamarckian kind, although some biologists, and philosophers such as Sir Karl Popper (Popper, 1972, p. 229; Popper and Eccles, 1977, pp. 12–13), have argued for an 'active Darwinism' in which purposeful behaviour is admitted.

In contrast, mainstream science has an exclusive emphasis on deterministic causality. Consequently there are difficulties in accommodating the concept of purposive behaviour in its traditional mode of thought. If purpose has a cause, then the cause alone is seen as relevant. The idea of an uncaused purpose is either regarded as an impossibility, or it is ignored because such spontaneity is taken to mean that there is no mechanism to be explained. Even if we regard this aspect of modern science as limited and inapplicable to social science, it still applies to the Darwinian idea of evolution through natural selection. In this, the existence of units or characteristics is explained by direct or indirect reference to survival values. If organisms have plans or intentions then they are not recognised; individual choices and purposes have no place in the explanation.

There is some truth in Veblen's depiction of the 'Darwinian scheme of thought' in terms of 'a continuity of cause and effect' (Veblen, [1907], 1919, pp. 436). However, he does not seem to wish to deal with the apparent exclusion of purposeful behaviour from such a 'mechanical' scheme. He is not led to adopt explicitly a more Lamarckian view of evolution, in which some notion of purposeful behaviour is admitted. Unlike Spencer, Veblen described himself as a Darwinian, partly because he puts greater stress on the selection process through which some institutions prosper and others became extinct.

Nevertheless, he simultaneously adopted the Lamarckian principle of the inheritance of acquired characters. Indeed, in one passage, writing after a discussion of the fixing of certain social traits, he seems indifferent between a Lamarckian and a Darwinian view of evolution: 'The point is not seriously affected by any question as to whether it was a process of habituation in the old-fashioned sense of the word or a process of selective adaptation of the race' (Veblen, 1899, p. 221).

Again there is insufficient clarity in Veblen's writing. There is a basis for limited mitigation, however. First, as Stephen Edgell and Rick Tilman (1989, p. 1009) point out, Lamarckian theory was still widely accepted, long after the publication of Darwin's *Origin*. Second, the scientific account of the mechanisms of the evolutionary process was

not altogether clear in Veblen's time, and around 1900 the Darwinians were somewhat in disarray. Third, even Darwin himself had flirted with elements of Lamarckism. As Peter Bowler (1988, p. 98) puts it: 'Darwin himself did not deny a limited role for the inheritance of acquired characters, and he was thus able to admit that the learning of new habits by the animals themselves can play a role.'

However, Darwin dallied with Lamarckism mainly in regard to the acceptance of the possibility of the inheritance of acquired characters. He did not adopt the whole Lamarckian explanation of variety and its sources, nor the central causal role of intentionality and purposeful behaviour. Veblen accepted the possibility of the inheritance of acquired characters, and without much else of the Lamarckian baggage, but made explicit reference to purposeful behaviour. In any case, the full-blooded Lamarckism of the sort adopted and developed by Spencer was alien to Veblen's 'post-Darwinian' thought.

4. In conclusion: the Veblenian legacy

To some extent, the degree of imprecision in the terms in which Veblen's evolutionary ideas were expressed became an impediment to their theoretical development. In part, this imprecision and incompleteness stems from the limited development of evolutionary theory in biology at his time. But this cannot explain all the vagaries and omissions.

Compared with the integrated theoretical constructions of, say, Marx or Marshall, there is not a comparably systematic theory of industry, technology or the macroeconomy in Veblen's work. Consequently, the institutionalists had to look elsewhere for a basis for their policy recommendations. It is partly because Veblen addressed economic systems in such complex and dynamic terms that he fails to provide a systemic theory. To some extent, his achievement is limited by ambition itself.

However, despite some problems and inadequacies, on the whole Veblen was relatively successful in establishing the basis of a Darwinian economics. First, the principle of 'idle curiosity' became the ongoing source of variety or mutation in the evolutionary process. Second, the institution became the unit of relative stability and continuity through time, ensuring that much of the pattern and variety is passed on from one period to the next, so that selection has relatively stable units upon which to operate. Third, mechanisms are identified through which well-adapted institutions are imitated and replicated, and the worse-adapted become extinct: analogous to the 'struggle for existence'.

Hence a principal component of this achievement is its embodiment of the idea of the cumulatively self-reinforcing institution as a unit of evolutionary selection, to be subject to the procedures of mutation and selection. For Veblen, the objects of economic evolution and selection are institutions and routines. Their fundamental genetic component is habits and instincts.

The nature of the evolutionary process governing these elements is selective rather than purely developmental, and phylogenetic rather than simply ontogenetic. Veblenian economic evolution, in other words, is not confined to the development of the organism from its genetic rules.¹ It is phylogenetic in that the ongoing processes of selection and development of the whole population of institutions is considered. His evolutionary theory is thus more extensive than that of Adam Smith or Carl Menger, and it does not share the exclusive biological reductionism to be found in the works of Spencer. Hence, despite

¹ The fact that the Veblenian cumulative process of evolutionary change of, say, consumer tastes, is phylogenetic, and not ontogenetic, was perhaps first noted by Copeland (1958, pp. 64, 67).

their limitations, Veblen's writings stand out as the most successful attempts, at least until the 1970s, to incorporate post-Darwinian biological thinking into economics and social science.

A particular relevance of Veblen's evolutionary theory is his analysis of science and technology in modern capitalism. It is clear from Veblen's writings that he regarded science and technology as one of the major motors of economic and social advance. The relevance of evolutionary principles from biology to the theory of technological change has again been recognised in recent years, with the consequence that many of Veblen's insights are being repeated and rediscovered, often without recognition of their precedent.¹

It is also apparent that Veblen kept largely up to date with developments in theoretical biology and was keen to apply relevant insights to economics and social science. There have been enormous developments in biology since his day, but economists have largely neglected his example and failed to attempt to incorporate the appropriate insights. We act in the spirit, if not the letter, of no greater economist than Veblen in attempting to draw inspiration from the science of life.

Veblen fathered American institutionalism but, probably owing to the limited development of biology in the early decades of the 20th century, in fact this tradition drew little from biology after Veblen. John Commons (1934, p. 45), for instance, clearly perceived institutional evolution in Veblenian terms.² However, his prime metaphor for his own core idea of a 'transaction' as a fundamental 'unit of activity'—his alternative to the abstract individual of neoclassical economics—was found not in biology but in quantum physics. The quantum was seen as a non-atomistic and analogous 'unit of activity' in which matter and energy, being and becoming, are all combined (*ibid.*, pp. 55–56). Generally, after Veblen, the biological metaphor lies fallow for many years, somewhat neglected even by the American institutionalists, used often in some loose sense where evolution is identified simply with development.³

Other major institutionalist writers, such as Clarence E. Ayres, also became excited by Darwinism and its potential application to social science.⁴ Prompted by Veblen's matter-of-fact view of science and technology, Ayres invested Veblen's concept of 'idle curiosity' with the full merits of human intelligence and technological progress (Rutherford, 1981). In contrast, incongruously distinguished from technology and science, institutional behaviour as regarded as ossified and 'ceremonial' by Ayres. One was given a positive charge, the other a negative (McFarland, 1986).

Apart from the problematic nature of the theoretical distinction between institutions and technology (Dosi, 1988; Samuels, 1977), Ayres failed to note the positive and enabling functions of institutional rigidity (Giddens, 1984; Hodgson, 1988). In this respect he differs from Commons (1934, p. 73), who clearly saw institutions as a liberating as well as a constraining force. Furthermore, it should be stressed that modern evolutionary theory

¹ Dosi's (1988) rich discussion of technological innovation has a strong Veblenian tone. Recent prominent discussions of technological evolution which briefly acknowledge Veblen include Clark and Juma (1987) and David (1985).

² See the discussions in Rutherford (1983) and Biddle (1990).

³ It is probably thus in part an accident that the strong revival in evolutionary theorising in economics since the 1970s is frequently described as 'Schumpeterian', rather than 'institutionalist' or 'Veblenian'. As noted elsewhere (Hodgson, 1993) the ideas that have emerged with the 'new wave' of evolutionary theorising have much more to do with Veblen and the 'old' institutionalism than with Schumpeter himself.

⁴ Ayres (1932) wrote a popular and 'racy' book on that champion of Darwinism and scourge of the creationists, Thomas Henry Huxley. Yet this book is more concerned with Huxley's scientific campaign to establish the fact of human descent from the apes, rather than the precise mechanisms of natural selection.

cannot assign an unambiguously negative sign to either error or rigidity, just as it cannot always regard even a hopeful mutation as positive. Evolution proceeds, not in a mechanical manner, of force against resistance, but through the *combination* of rigid and durable entities—genes or institutions—with unceasing selection and change.

Despite its limitations, Veblen's understanding of the nature of evolution was more acute and his ambitions for an evolutionary economics were more substantial. Richard Hofstadter's (1959, pp. 152–155) remarks here are appropriate:

Veblen's conception of the uses of Darwinism in economics was not the most representative of his generation, but in the long run it may be the most enduring . . . Where other economists had found in Darwinian science merely a source of plausible analogies or a fresh rhetoric to substantiate traditional postulates and precepts, Veblen saw it as a loom upon which the whole fabric of economic thinking could be rewoven.

In his relative success with the evolutionary metaphor, Veblen speaks more loudly and clearly than Marx, although his theoretical system as a whole lacked the latter's symphonic grandeur (Hodgson, 1992). Brinley Thomas (1991) shows that Marshall's adoption of evolutionary ideas from biology was more promise than substance, and more Spencerian than Darwinian. It is argued elsewhere (Hodgson, 1993), with an equivalent degree of heresy, that Veblen's use of evolutionary thinking from biology was much more extensive than that of Schumpeter. Veblen should thus be placed amongst the founding figures of modern evolutionary economics: perhaps even the greatest of them all.

Bibliography

- Arthur, W. B. 1989. Competing technologies, increasing returns, and lock-in by historical events, *The Economic Journal*, vol. 99, no. 1, March
- Arthur, W. B. 1990. Positive feedbacks in the economy, *Scientific American*, vol. 262, no. 2, February
- Ayres, C. E. 1932. *Huxley*, New York, Norton
- Ayres, C. E. 1944. *The Theory of Economic Progress*, 1st edn, Chapel Hill, North Carolina, University of North Carolina Press
- Biddle, J. E. 1990. Purpose and evolution in Commons's institutionalism, *History of Political Economy*, vol. 22, no. 1, Spring
- Bowler, P. J. 1983. *The Eclipse of Darwinism: Anti-Darwinian Evolution Theories in the Decades Around 1900*, Baltimore, Johns Hopkins University Press
- Bowler, P. J. 1988. *The Non-Darwinian Revolution: Reinterpreting a Historical Myth*, Baltimore, Johns Hopkins University Press
- Burkhardt Jr, R. W. 1977. *The Spirit of the System: Lamarck and Evolutionary Biology*, Cambridge, Mass., Harvard University Press
- Chasse, J. D. 1984. Marshall, the human agent and economic growth: wants and activities revisited, *History of Political Economy*, vol. 16, no. 3, Fall
- Clark, N. G. and Juma, C. 1987. *Long-Run Economics: an Evolutionary Approach to Economic Growth*, London, Pinter
- Commons, J. R. 1934. *Institutional Economics—its Place in Political Economy*, New York: Macmillan, reprinted with a new introduction by M. Rutherford, New Brunswick, Transaction Publishers (1990)
- Copeland, M. A. 1958. On the scope and method of economics, in Dowd, D. F. (1958)
- David, P. A. 1985. Clio and the economics of QWERTY, *American Economic Review*, vol. 75, no. 2, May
- Dorfman, J. 1934. *Thorstein Veblen and His America*, New York, Viking Press
- Dorfman, J. 1964. Introductory note, to 1964 edition of Veblen, T. B. (1914)
- Dosi, G. 1988. The sources, procedures, and microeconomic effects of innovation, *Journal of Economic Literature*, vol. 26, no. 3, September
- Dowd, D. F. (ed.) 1958. *Thorstein Veblen: a Critical Appraisal*, Ithica, N.Y., Cornell University Press

- Dyer, A. W. 1984. The habit of work: a theoretical exploration, *Journal of Economic Issues*, vol. 18, no. 2, June
- Dyer, A. W. 1986. Veblen on scientific creativity, *Journal of Economic Issues*, vol. 20, no. 1, March
- Edgell, S. 1975. Thorstein Veblen's theory of evolutionary change, *American Journal of Economics and Sociology*, vol. 34, July
- Edgell, S. and Tilman, R. 1989. The intellectual antecedents of Thorstein Veblen: a reappraisal, *Journal of Economic Issues*, vol. 23, no. 4, December
- Eff, E. A. 1989. History of thought as ceremonial genealogy: the neglected influence of Herbert Spencer on Thorstein Veblen, *Journal of Economic Issues*, vol. 23, no. 3, September
- Eldredge, N. and Gould, S. J. 1977. Punctuated equilibria: the tempo and mode of evolution reconsidered, *Paleobiology*, vol. 3
- Giddens, A. 1984. *The Constitution of Society: Outline of the Theory of Structuration*, Cambridge, Polity Press
- Gould, S. J. 1982. The meaning of punctuated equilibrium and its role in validating a hierarchical approach to macroevolution, in Milkman, R. (ed.) (1982), *Perspectives on Evolution*, Sunderland, Mass.: Sinauer Associates
- Hamilton, D. B. 1991. *Evolutionary Economics: A Study in Change in Economic Thought*, 3rd edn, New Brunswick, N.J., Transaction Publishers
- Hamilton, W. H. 1963. Institution, in Seligman, E. R. A. and Johnson, A. (eds.), *Encyclopaedia of the Social Sciences*, vol. 7, London, Macmillan
- Hobson, J. A. 1914. *Work and Wealth*, London, Macmillan
- Hodgson, G. M. 1988. *Economics and Institutions: A Manifesto for a Modern Institutional Economics*, Cambridge, Polity Press
- Hodgson, G. M. 1991A. Hayek's theory of cultural evolution: an evaluation in the light of Vanberg's critique, *Economics and Philosophy*, vol. 7, no. 1, March
- Hodgson, G. M. 1991B. Economic evolution: intervention contra Pangloss, *Journal of Economic Issues*, vol. 25, no. 2, June
- Hodgson, G. M. 1992. Marx, Engels and economic evolution, *International Journal of Social Economics*, forthcoming
- Hodgson, G. M. 1993. *Economics and Evolution*, Cambridge, Polity Press, forthcoming
- Hofstadter, R. 1959. *Social Darwinism in American Thought*, revised edn, New York, Braziller
- Hill, F. G. 1958. Veblen and Marx, in Dowd, D. F. (1958)
- Humphrey, T. M. 1990. Cumulative process models from Thornton to Wicksell, in Moggridge, D. E. (ed.), (1990), *Perspectives in the History of Economic Thought. Volume Four: Keynes, Macroeconomics and Method*, Aldershot, Edward Elgar
- Ingrao, B. and Israel, G. 1990. *The Invisible Hand: Economic Equilibrium in the History of Science*, Cambridge, Mass., MIT Press
- Jones, L. B. 1986. The institutionalists and 'On the Origin of Species': a case of mistaken identity, *Southern Economic Journal*, vol. 52, no. 4, April
- Kaldor, N. 1972. The irrelevance of equilibrium economics, *The Economic Journal*, vol. 82, no. 4, December, reprinted in Kaldor, N. (1978), *Further Essays on Economic Theory: (Collected Economic Essays Vol. 5)*, London, Duckworth, and in Targetti, F. and Thirlwall, A. P. (eds.) (1989), *The Essential Kaldor*, New York: Holmes and Meier
- Kapp, K. W. 1976. The nature and significance of institutional economics, *Kyklos*, vol. 29, reprinted in Samuels, W. J. (ed.) (1988), *Institutional Economics*, vol. 2, Aldershot, Edward Elgar
- Knight, F. H. 1935. Intellectual confusion on morals and economics, *International Journal of Ethics*, vol. 45, no. 1, January
- Langlois, R. N. 1989. What was wrong with the old institutional economics (and what is still wrong with the new), *Review of Political Economy*, vol. 1, no. 3, November
- Leathers, C. G. 1990. Veblen and Hayek on instincts and evolution, *Journal of the History of Economic Thought*, vol. 12, no. 2, June
- Marshall, A. 1949. *The Principles of Economics*, 8th (reset) edn, London, Macmillan
- Marx, K. 1977. *Karl Marx: Selected Writings* (edited by D. McLellan), Oxford, Oxford University Press
- Mayhew, A. 1987. The beginnings of institutionalism, *Journal of Economic Issues*, vol. 21, no. 3, September
- Mayr, E. 1982. *The Growth of Biological Thought*, Cambridge, Mass., Harvard University Press

- McDougall, W. 1909. *An Introduction to Social Psychology*, Boston, Luce
- McFarland, F. 1986. Clarence Ayres and his gospel of technology, *History of Political Economy*, vol. 18, no. 4, Winter
- Mirowski, P. 1989. *More Heat Than Light: Economics as Social Physics, Physics as Nature's Economics*, Cambridge, CUP
- Mokyr, J. 1990. *The Lever of Riches: Technological Creativity and Economic Progress*, Oxford, Oxford University Press
- Mokyr, J. (forthcoming). Evolutionary biology, technical change, and economic history, *Bulletin of Economic Research*
- Murphree, I. L. 1959. Darwinism in Thorstein Veblen's economics, *Social Research*, vol. 26, no. 2, June
- Myrdal, G. 1939. *Monetary Equilibrium*, London, Hodge
- Myrdal, G. 1944. *An American Dilemma: The Negro Problem and Modern Democracy*, New York, Harper & Row
- Myrdal, G. 1957. *Economic Theory and Underdeveloped Regions*, London, Duckworth
- Myrdal, G. 1978. Institutional economics, *Journal of Economic Issues*, vol. 12, no. 4, December
- Nelson, R. R. and Winter, S. G. 1982. *An Evolutionary Theory of Economic Change*, Cambridge, Mass., Harvard University Press
- Nitecki, M. H. (ed.) 1988. *Evolutionary Progress*, Chicago, University of Chicago Press
- Parsons, T. 1937. *The Structure of Social Action*, New York, McGraw-Hill
- Peel, J. D. Y. 1971. *Herbert Spencer: the Evolution of a Sociologist*, New York, Basic Books
- Peirce, C. S. 1878. How to make our ideas clear, *Popular Science Monthly*, vol. 12, January, reprinted in Peirce, C. S. (1934), *Collected Papers of Charles Sanders Peirce, Volume V, Pragmatism and Pragmaticism*, edited by C. Hartshorne and P. Weiss, Cambridge, Mass., Harvard University Press
- Popper, K. R. 1972. *Objective Knowledge: an Evolutionary Approach*, Oxford, Oxford University Press
- Popper, K. R. and Eccles, J. C. 1977. *The Self and Its Brain*, Berlin, Springer International
- Riesman, D. 1963. *Thorstein Veblen: a Critical Interpretation*, New York, Charles Scribner's Sons
- Rutherford, M. 1981. Clarence Ayres and the instrumentalist theory of value, *Journal of Economic Issues*, vol. 15, no. 3, September, reprinted in Samuels, W. J. (ed.) (1988), *Institutional Economics*, vol. 3, Aldershot, Edward Elgar
- Rutherford, M. 1983. J. R. Commons's institutional economics, *Journal of Economic Issues*, vol. 17, no. 3, September, reprinted in Samuels, W. J. (ed.) (1988), *Institutional Economics*, vol. 1, Aldershot, Edward Elgar
- Rutherford, M. 1984. Thorstein Veblen and the processes of institutional change, *History of Political Economy*, vol. 16, no. 3, Fall
- Samuels, W. J. 1977. Technology vis-à-vis institutions in the JEI: a suggested interpretation, *Journal of Economic Issues*, vol. 11, no. 4, December, reprinted in Samuels, W. J. (ed.) (1988), *Institutional Economics*, vol. 2, Aldershot, Edward Elgar
- Schumpeter, J. A. 1976. *Capitalism, Socialism and Democracy*, 5th edn, London, George Allen & Unwin
- Seckler, 1975. *Thorstein Veblen and the Institutionalists: A Study in the Social Philosophy of Economics*, London, Macmillan
- Somit, A. and Peterson, S. A. 1989. The punctuated equilibrium debate: scientific issues and implications, *Journal of Social Biology and Structuration*, vol. 12, nos. 2/3
- Spencer, H. 1855. *The Principles of Psychology*, London, Williams and Norgate
- Thomas, B. 1991. Alfred Marshall on economic biology, *Review of Political Economy*, vol. 3, no. 1, January
- Tilman, R. 1987. Some recent interpretations of Thorstein Veblen's theory of institutional change, *Journal of Economic Issues*, vol. 21, no. 2, June
- Veblen, T. B. 1892. Some neglected points in the theory of socialism, *Annals of the American Academy of Political and Social Science*, vol. 2, reprinted in Veblen, T. B. (1919)
- Veblen, T. B. 1898A. Why is economics not an evolutionary science?, *Quarterly Journal of Economics*, vol. 12, reprinted in Veblen, T. B. (1919)
- Veblen, T. B. 1898B. The instinct of workmanship and the irksomeness of labor, *American Journal of Sociology*, vol. 4, September, reprinted in Veblen, T. B. (1934)

- Veblen, T. B. 1899. *The Theory of the Leisure Class: An Economic Study of Institutions*, New York, Macmillan
- Veblen, T. B. 1900. The preconceptions of economic science: III, *Quarterly Journal of Economics*, vol. 14, no. 1, February, reprinted in Veblen (1919)
- Veblen, T. B. 1904. *The Theory of Business Enterprise*, New York, Charles Scribners Sons, reprinted by Augustus Kelley
- Veblen, T. B. 1907. The socialist economics of Karl Marx and his followers II: the later Marxism, *Quarterly Journal of Economics*, vol. 21, no. 1, February, reprinted in Veblen (1919)
- Veblen, T. B. 1908A. Professor Clark's economics, *The Quarterly Journal of Economics*, vol. 22, no. 1, February
- Veblen, T. B. 1908B. The evolution of the scientific point of view, *University of California Chronicle*, vol. 10, no. 4, reprinted in Veblen (1919)
- Veblen, T. B. 1909. The limitations of marginal utility, *Journal of Political Economy*, vol. 17, reprinted in Veblen (1919)
- Veblen, T. B. 1914. *The Instinct of Workmanship, and the State of the Industrial Arts*, New York: Augustus Kelley, reprinted with a new introduction by M. G. Murphey and a 1964 introductory note by J. Dorfman, New Brunswick, Transaction Books (1990)
- Veblen, T. B. 1915. *Imperial Germany and the Industrial Revolution*, New York, Macmillan
- Veblen, T. B. 1919. *The Place of Science in Modern Civilization and Other Essays*, New York, Huebsch, reprinted with a new introduction by W. J. Samuels, New Brunswick, Transaction Publishers (1990)
- Veblen, T. B. 1925. Economic theory in the calculable future, *American Economic Review*, vol. 15, no. 1, Supplement, March, reprinted in Veblen, T. B. (1934)
- Veblen, T. B. 1934. *Essays on Our Changing Order*, ed. L. Ardzrooni, New York, The Viking Press
- Walker, D. A. 1977. Thorstein Veblen's economic system, *Economic Inquiry*, vol. 15, no. 2, April, reprinted in Samuels, W. J. (ed.) (1988), *Institutional Economics*, vol. 1, Aldershot, Edward Elgar
- Young, A. A. 1928. Increasing returns and economic progress, *The Economic Journal*, vol. 38, no. 4, December
- Young, R. M. 1969. Malthus and the evolutionists: the common context of biological and social theory, *Past and Present*, no. 43, May